

Developing Tools to Identify Ancient Grasslands and Understanding Their Value

Background

Much like in ancient woodlands, continuity of management and absence of significant disturbance results in the accrual of value within grasslands in the form of:

- Carbon storage in soil
- Diverse soil microbial communities
- Greater richness of invertebrates
- High occurrence of less common, rare, scarce or declining plants
- Potentially more genetically resilient populations



Although the rate at which grasslands acquire these features may vary, it is likely that there is a threshold of ecological maturity where their value comes significantly higher than that of recently restored or newly created neutral grasslands. However, ancient grasslands are not currently protected as “Irreplaceable Habitats”, unlike ancient woodlands. This may be due to the lack of a formal definition and reliable tools to identify them.

Recent work commissioned by **Plantlife**, produced by the **British Ecological Society** (BES) and developed with input from 22 expert consultees explores how ‘ancient grasslands’ might be defined in practice. The full report is available to read here: [Discussion Paper on Ancient Grasslands](#)

The paper evaluates various ways to identify ancient grasslands, including the use of:

- Vascular plant assemblages
- CHEGD fungi
- Invertebrate communities
- Soil properties

These insights provide an opportunity to develop a region-specific list of indicator species and community traits for identifying ancient grasslands in the UK for the first time.

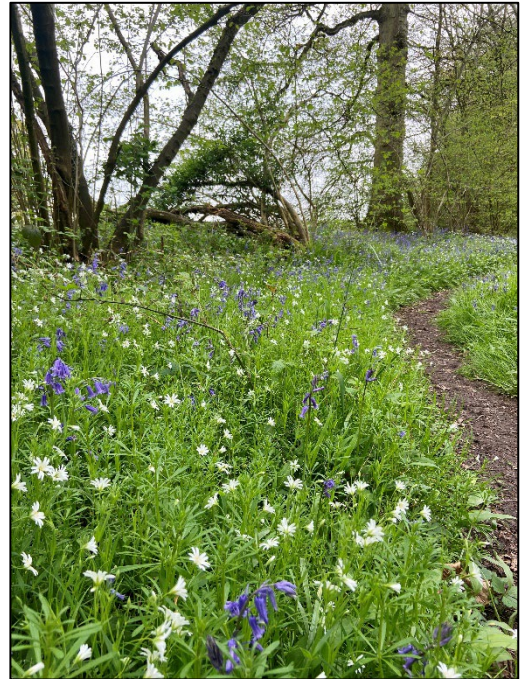


Rationale

Indicator species and plant community traits are essential tools for classifying habitats. They underpin widely used classification systems such as **UKHabs**, **NVC**, and **Phase 1**, which inform both planning and conservation efforts. In woodland conservation, **regional indicator species lists** have been instrumental in identifying and protecting ancient woodlands.

At present, however, no tested list exists for **lowland neutral grasslands**, and the composition of ancient grassland indicator species is also highly site-specific or geographically isolated, which increases the risk of inbreeding due to limited genetic connectivity across landscapes.

Worcestershire is a stronghold for lowland neutral grasslands with continuity of management and therefore is an appropriate starting county for establishing a regional ancient grassland indicator species list. Furthermore, the history of management at many sites is well documented and the distribution of species-rich neutral grasslands well mapped and is estimated to host **7.55% of England's lowland meadows** (WWT, 2023).



Aims

1. Develop an **ancient grassland indicator species list** specific to Worcestershire (VC37).
2. Assess **community composition traits** of ancient grasslands in the region and how these change over time.
3. Examine **genetic connectivity** and diversity in ancient grassland indicator species.
4. Analyse **soil characteristics** such as pH, nutrient levels, and microbial activity.



This Year

Vegetation Survey

This summer, I will survey around 20 neutral grasslands of varying age and size using the methods outlined in “Field_Survey_Protocol_2025”. The aim is to record plant species composition, identify likely indicator species, and document additional features (e.g. ant hill abundance, ridge and furrow presence) that may help signal whether a site is likely ‘ancient.’

Genetic Diversity

To explore genetic connectivity and diversity, I will collect leaf¹ samples from a few key indicator species:

- *Genista tinctoria* (Dyer’s greenweed)
- *Colchicum autumnale* (Meadow saffron)
- *Carex caryophyllea* (Spring sedge)
- *Ophioglossum vulgatum* (Adder’s tongue)
- *Serratula tinctoria* (Saw-wort)

Next Year

Soil Analysis

Soil samples will be collected in **late winter to early spring (Feb–April 2026)** to assess nutrient levels, pH, and microbial activity. This timing avoids disturbing plant growth and ensures conditions are suitable for measuring baseline soil traits. Soil sampling will only take place where landowner permission has been granted.

Thank You

Thank you very much for your interest and support.

¹ Samples will consist of a small number of individual leaves or a small section of an individual leaf depending upon the growth form of the species. Sampling is avoided where the abundance of species at a particular site is low. See [here](https://doi.org/10.1007/s10531-019-01811-x) for a study which uses the methods this study aims to implement (DOI: <https://doi.org/10.1007/s10531-019-01811-x>)

